

A Narrowing Geometry in Contaminated Reasoning

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Background: Data Contamination

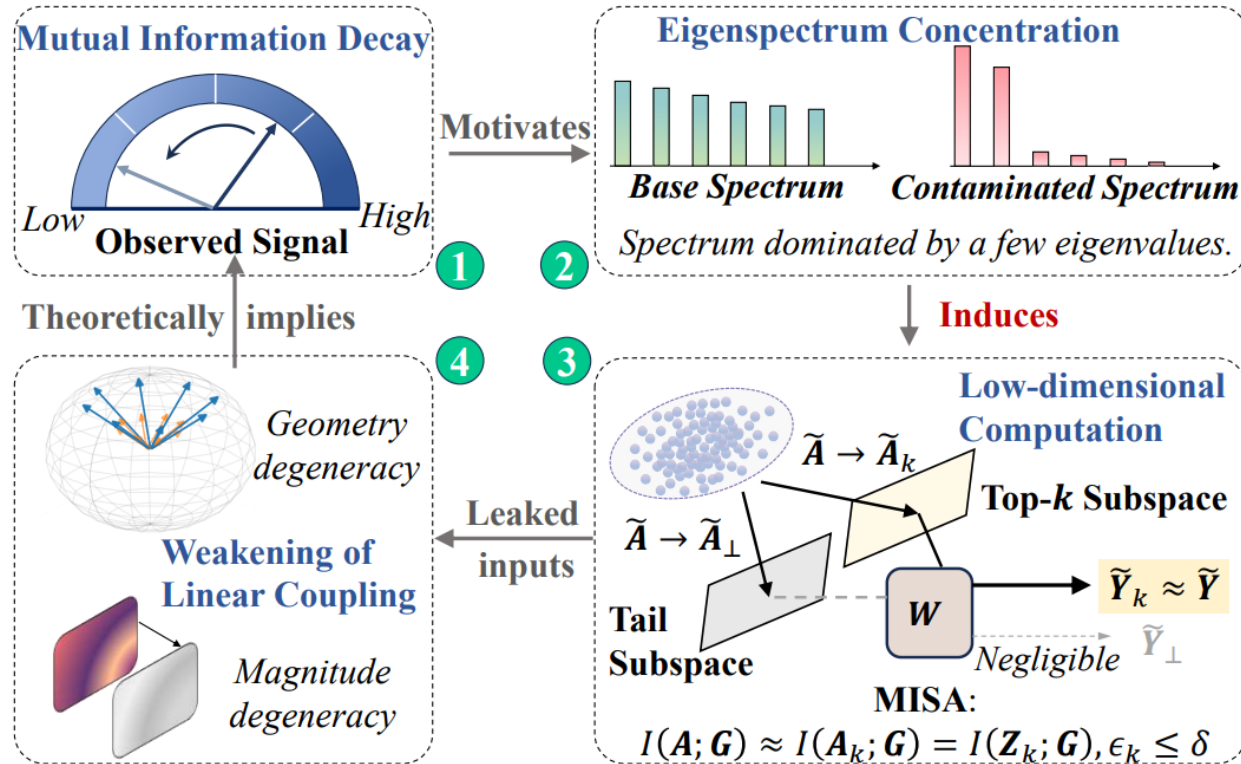
■ Motivation

- **Benchmark contamination is widespread.**
Leaked data can induce *contaminated reasoning*.
- **Strong performance can be misleading.**
Does high accuracy imply genuine reasoning?
- **Robust evaluation remains difficult.**
It is hard to design reliable evaluation framework.

■ Key Question

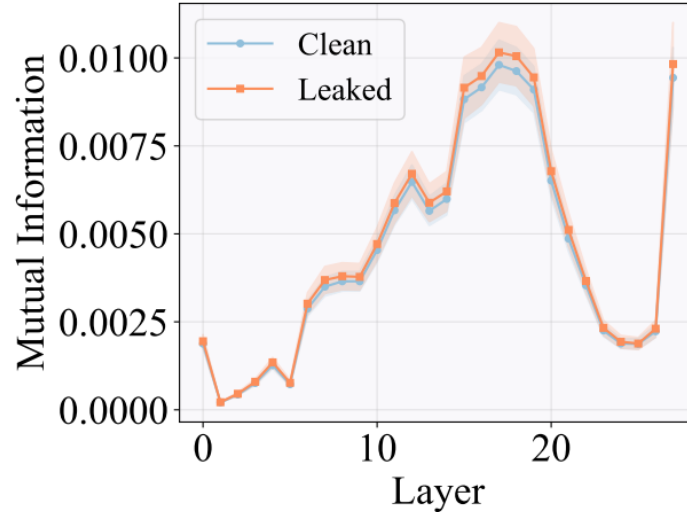
What changes inside the model when reasoning is contaminated?

Overview

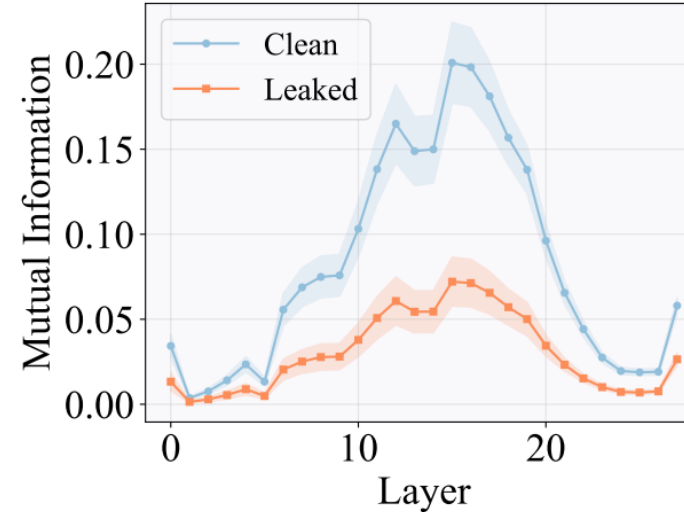


- **Signal:** Mutual Information Decay
- **Mechanism:** Eigenspectrum Concentration → Low-dimensional Computation
- **Explanation:** Why Mutual Information Decay happens
- **Application:** Reasoning Restoration

Mutual Information Decay



(a) Base Model



(b) Contaminated Model

■ Metric: Gaussian Mutual Information

$$I(\mathbf{A}; \mathbf{G}) = \frac{1}{2} \log \frac{\det(\mathbf{C}_{\mathbf{A}}) \det(\mathbf{C}_{\mathbf{G}})}{\det(\mathbf{C})}$$

■ Observation

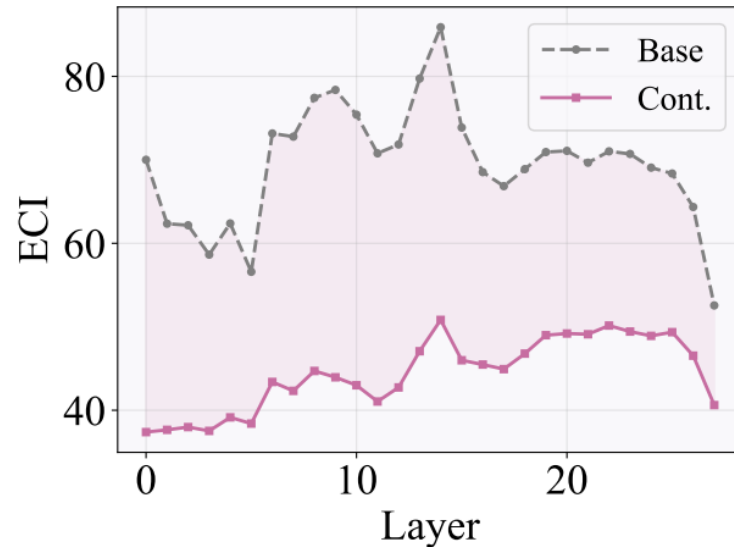
- Base model: $\text{Leaked} \approx \text{Clean}$
- Contaminated model: $\text{Leaked} < \text{Clean}$

■ Takeaway

- Mutual information decay is a distinctive signal of contaminated reasoning.

Mechanisms

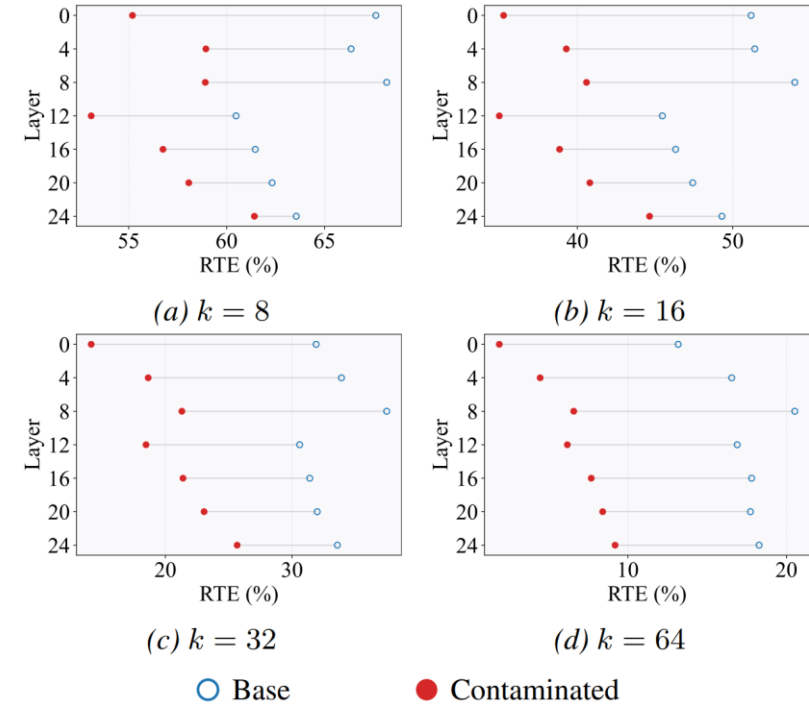
■ Eigenspectrum Concentration



■ Takeaways

- Lower ECI $\rightarrow$ variance dominated by a few eigenvalues.
- Contaminated activations exhibit stronger eigenspectrum concentration.

■ Low-Dimensional Computation



■ Takeaways

- Lower RTE $\rightarrow$ stronger low-dimensional computation.
- Module outputs are well captured by a small top- k subspace.

Causal Validation

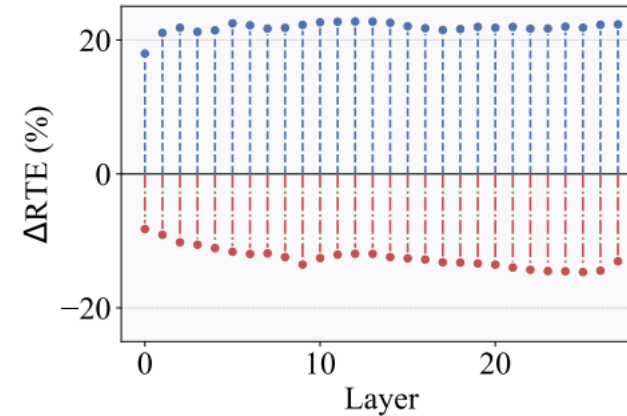
■ No Significant Tail Vector Amplification

MODELS	METRICS	k			
		8	16	32	64
BASE	R_{worst}	1.82	1.80	1.78	1.76
	R_{med}	0.93	0.94	0.95	0.98
CONTAMINATED	R_{worst}	1.71	1.69	1.68	1.60
	R_{med}	0.95	0.95	0.96	0.98

■ Observation

- Transformed tail eigenvectors have comparable norms to principal eigenvectors.

■ Spectrum Reshaping



Flatten: RTE ↓

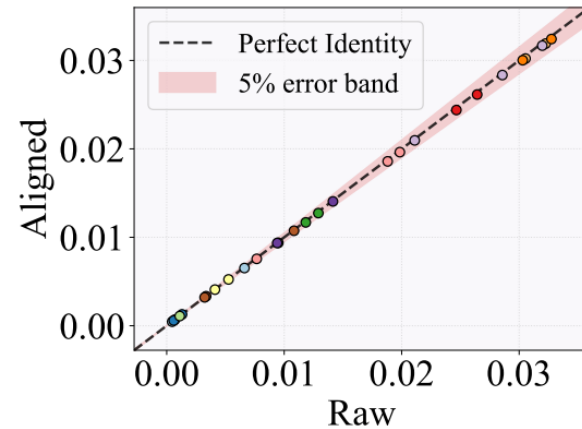
Sharpen: RTE ↑

■ Observation

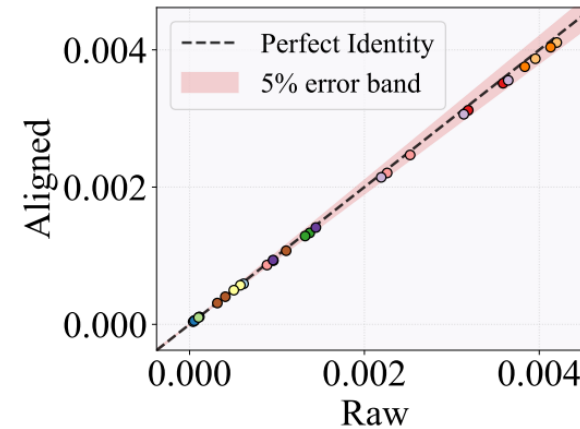
- Manipulating only the eigenspectrum changes the computation dimensionality.

Explanation of Mutual Information Decay

■ Mutual Information Subspace Alignment (MISA)



(a) Clean Data



(b) Leaked Data

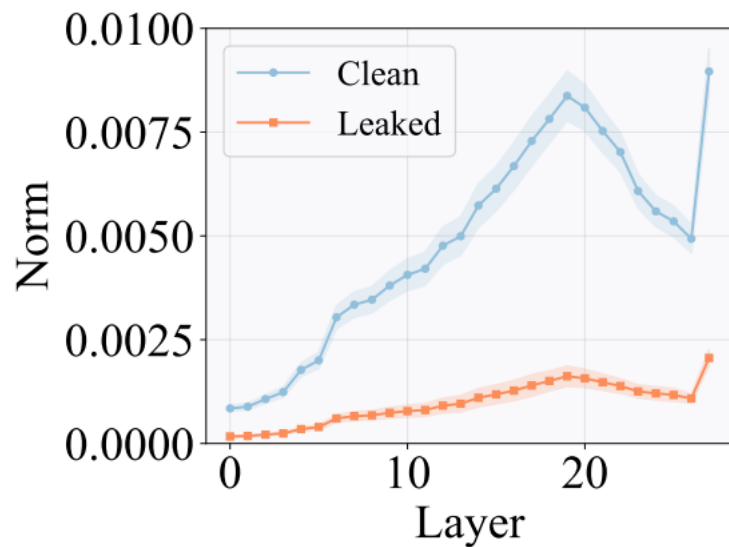
$$I(A; G) \approx I(A_k; G) = I(Z_k; G) \quad (\epsilon_k \leq \delta)$$

■ Takeaway

- The mutual information is determined primarily within the active computation subspace.

Explanation of Mutual Information Decay

■ Weakening of Linear Coupling

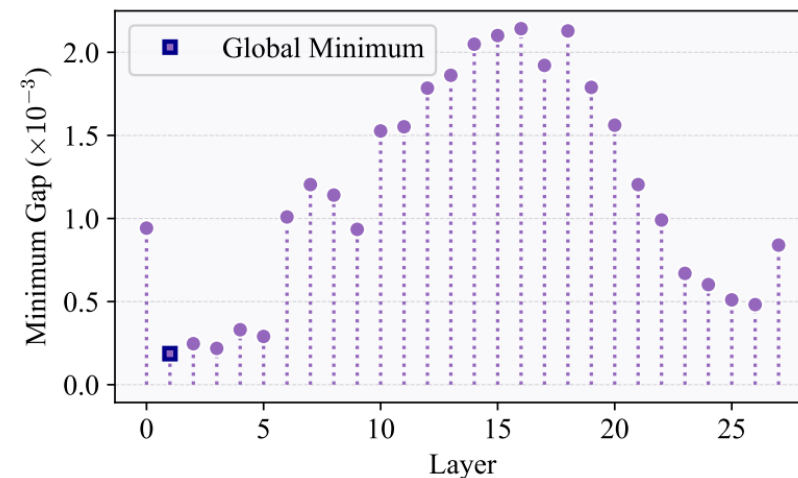


$$\left\| C_{Z_k G}^{(l)} \right\|_F < \left\| C_{Z_k G}^{(c)} \right\|_F$$

■ Observation

- Leaked data shows weaker representation-gradient coupling in the active subspace.

■ Singular Value Partial Order



$$R = C_{Z_k}^{-\frac{1}{2}} C_{Z_k G} C_G^{-\frac{1}{2}} \quad \rho_i^{(l)} \leq \rho_i^{(c)} \quad \forall i, \quad \text{and} \quad \exists j : \rho_j^{(l)} < \rho_j^{(c)}$$

■ Observation

- The decay holds direction-wise in the whitened canonical space.

Application: Reasoning Restoration

■ Method: Top-Subspace Suppression (TSS)

$$\mathbf{a}^{\text{new}} = \boldsymbol{\mu} + \left(\mathbf{I} - \gamma^\alpha \mathbf{U}_k \mathbf{U}_k^\top \right) (\mathbf{a} - \boldsymbol{\mu}), \quad \gamma = \frac{\left\| \mathbf{U}_k \mathbf{U}_k^\top (\mathbf{a} - \boldsymbol{\mu}) \right\|}{\left\| \mathbf{a} - \boldsymbol{\mu} \right\|}, \quad \alpha \in [0, 1].$$

■ Quantitative Results

MODELS	STRATEGY	RESTOR. (↑)	RETAIN (↑)	AVG. (↑)
QWEN-1.5B	CONT.	0.00	100.00	50.00
	SAMPLING	23.08	89.02	56.05
	TED	27.47	86.83	57.15
	LNE-B.	39.56	89.96	64.76
	TSS (OURS)	51.65	94.37	73.01
QWEN-7B	CONT.	0.00	100.00	50.00
	SAMPLING	9.09	91.57	50.33
	TED	22.73	89.33	56.03
	LNE-B.	31.82	88.76	60.29
	TSS (OURS)	45.45	97.19	71.32

- TSS achieves the best average consistency.

■ Qualitative Case

Question

Ground Truth: $50000 \times 0.01 \times 0.05 = 25$

Marcy makes homemade candles that she markets as 99% guaranteed not to explode. 5% of the more dangerous candles also have a defect that makes them smell like wet dog. If she makes 50000 candles, how many of them will both smell like wet dog and explode?

Base Model (✗)

... Marcy makes 50,000 candles... 99% of these...
 $0.99 \times 50,000 = 49,500$ candles...
5% of these 49,500 candles have defects.
 $0.05 \times 49,500 = 2,475$ defective candles...
Thus, the number of candles that both smell like wet dog and explode is: **2475**

TSS + Contaminated Model (✗)

... find the number of candles that explode: $.99 * 50000 = 49500$ candles...
Then multiply that number of candles with the other defect to find the total number of them: $49500 * .05 = 2475$ candles... **2475**

- TSS makes the contaminated model follow the base model's reasoning trajectory.

Mechanistic understanding enables reasoning restoration.

Summary

■ Signal

- Mutual Information Decay

■ Mechanism

- Narrowing Geometry

■ Application

- Reasoning Restoration

THANK YOU